

# **BIG DATA ANALYTICS – CASE STUDY**

## **MEDTRACK – HADOOP MAPREDUCE ANALYTICS**

### **TEAM - 3**

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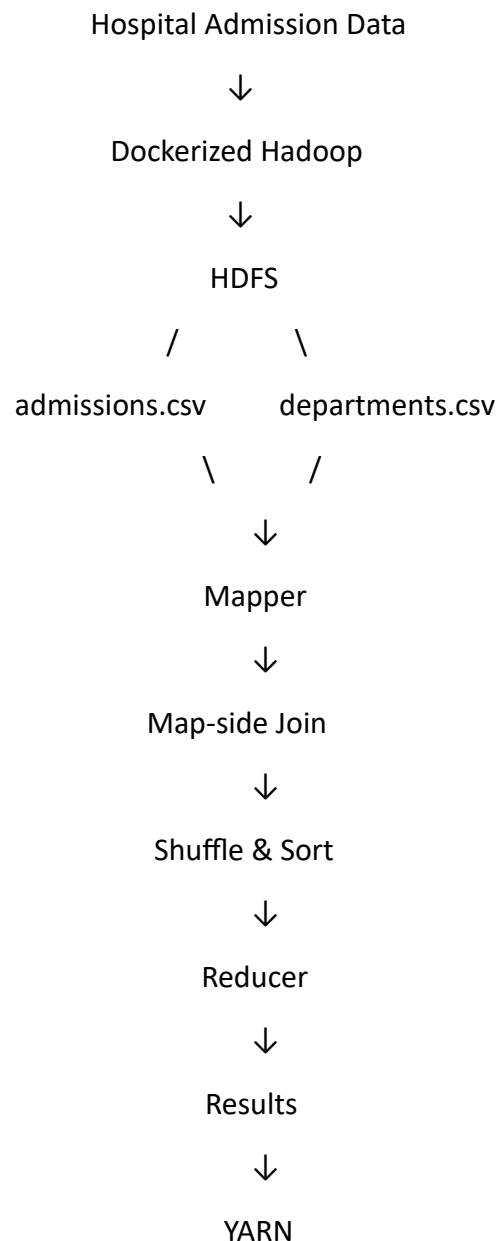
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## Introduction

MedTrack is a healthcare data analytics project implemented using Hadoop MapReduce. The project processes hospital admission data stored in HDFS and performs five analytical tasks related to department admissions, patient stay, monthly turnover, department heads, and monthly admission trends. The MapReduce jobs are executed in a Dockerized Hadoop environment using HDFS and YARN.

## System Architecture



The admission and department datasets are stored in HDFS. The Mapper processes admission records and, where required, uses the department dataset through a Map-side Join. Hadoop performs Shuffle and Sort to group intermediate results. The Reducer performs

the required aggregation and generates the final output. YARN manages the execution of the MapReduce jobs.

## Hadoop Environment

| Component           | Technology       |
|---------------------|------------------|
| Operating System    | Ubuntu/WSL       |
| Container Platform  | Docker           |
| Processing          | Hadoop MapReduce |
| Storage             | HDFS             |
| Resource Management | YARN             |
| Hadoop Version      | Hadoop 3.2.1     |

## HDFS Data Ingestion

```
hdfs dfs -ls /medtrack/input
```

### OUTPUT:

```
root@685e756c510d:/tmp# hdfs dfs -ls /medtrack/input
Found 2 items
-rw-r--r--  3 root supergroup      10976 2026-09-21 16:16 /medtrack/input/admissions.csv
-rw-r--r--  3 root supergroup       359 2026-09-21 16:21 /medtrack/input/departments.csv
```

## Problem 1 – Total Admissions per Department

### Objective

Calculate the total number of admissions/patients for each department.

### Approach

The Mapper uses a Map-side Join to obtain the department name from the department dataset and emits the department name with a count of 1. The Reducer sums the counts for each department.

### OUTPUT:

```
PS C:\Users\B Lalith kumar\Desktop\MedTrack-Problem1> docker exec namenode hdfs dfs -cat
/output/medtrack-problem1/part-r-00000
WARNING: log4j.properties is not found. HADOOP_CONF_DIR may be incomplete.
Cardiology      4
Neurology       3
Orthopedics     3
PS C:\Users\B Lalith kumar\Desktop\MedTrack-Problem1>
```

## Problem 2 – Average Length of Stay

### Objective

Calculate the average length of stay for each department.

## Approach

The Mapper calculates the length of stay using the admission and discharge dates and emits the department with the calculated stay. The Reducer aggregates the stay values and calculates the average.

## OUTPUT:

```
root@165f0de63dbe:~/medtrack# hdfs dfs -cat /medtrack/output_p2/part-r-00000
2026-09-27 17:10:13,409 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteHostTrusted = false
Cardiology      : 4.66 days
Dermatology     : 5.00 days
ENT             : 4.05 days
General Medicine : 4.60 days
Gynecology      : 4.05 days
Neurology       : 4.43 days
Oncology        : 4.68 days
Orthopedics     : 4.29 days
Pediatrics      : 4.60 days
Pulmonology     : 4.55 days
```

## Problem 3 – Highest Patient Turnover in a Month

### Objective

Find the department with the highest patient turnover for each month.

### Approach

The Mapper extracts the admission month and department name. Hadoop groups records by month. The Reducer counts admissions for each department within each month and selects the department with the highest count.

## OUTPUT:

```
root@685e756c510d:/tmp# hdfs dfs -cat /med
2026-09-21 16:49:20,076 INFO sasl.SaslData
2026-01 : General Medicine (8 patients)
2026-02 : Neurology (7 patients)
2026-03 : Cardiology (7 patients)
2026-04 : General Medicine (8 patients)
2026-05 : Neurology (7 patients)
2026-06 : General Medicine (6 patients)
root@685e756c510d:/tmp#
```

## Problem 4 – Department Head with Most Admissions

### Objective

Identify the department head overseeing the highest number of admissions.

### Approach

The Mapper performs a Map-side Join using the department code to obtain the department head and emits the head with a count of 1. The Reducer sums the admissions for each head and identifies the head with the highest admissions.

OUTPUT:

```
root@2f9231d56341:/tmp# hdfs dfs -cat /user/medtrack/output/problem4/part-r-00000
2026-09-20 16:59:06,003 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteHostTrusted = false
Dr. Priya Nair 42
root@2f9231d56341:/tmp#
```

Problem 5 – Month-wise Hospital Admissions

Objective

Calculate the total number of hospital admissions for each month.

Approach

The Mapper extracts the month from the admission date and emits the month with a count of 1. The Reducer sums the counts for each month.

OUTPUT:

```
hadoop@namenode:/tmp$ hdfs dfs -cat /hospital/output/problem5/part-r-00000
2026-01 7
2026-02 8
2026-03 10
2026-04 5
```

YARN Resource Manager Screenshot:

Cluster

About

Nodes

Node Labels

Applications

NEW

NEW SAVING

SUBMITTED

ACCEPTED

RUNNING

FINISHED

FAILED

KILLED

Scheduler

Tools

Cluster Metrics

|                |  |              |  |              |  |                |  |                    |  |             |  |              |  |                 |  |
|----------------|--|--------------|--|--------------|--|----------------|--|--------------------|--|-------------|--|--------------|--|-----------------|--|
| Apps Submitted |  | Apps Pending |  | Apps Running |  | Apps Completed |  | Containers Running |  | Memory Used |  | Memory Total |  | Memory Reserved |  |
| 0              |  | 0            |  | 0            |  | 0              |  | 0                  |  | 0 B         |  | 16 GB        |  | 0 B             |  |

Cluster Nodes Metrics

|              |  |                       |  |                      |  |            |  |                 |  |
|--------------|--|-----------------------|--|----------------------|--|------------|--|-----------------|--|
| Active Nodes |  | Decommissioning Nodes |  | Decommissioned Nodes |  | Lost Nodes |  | Unhealthy Nodes |  |
| 1            |  | 0                     |  | 0                    |  | 0          |  | 0               |  |

Scheduler Metrics

|                    |  |                               |  |                         |  |                         |  |
|--------------------|--|-------------------------------|--|-------------------------|--|-------------------------|--|
| Scheduler Type     |  | Scheduling Resource Type      |  | Minimum Allocation      |  | Maximum Allocation      |  |
| Capacity Scheduler |  | [memory-mb (unit=Mi), vcores] |  | <memory:1024, vCores:1> |  | <memory:8192, vCores:4> |  |

Show 20 entries

| ID                             | User | Name                    | Application Type | Queue   | Application Priority | StartTime                      | LaunchTime                     | FinishTime                     | State    | FinalStatus | Running Containers | Allocated CPU VCores | Allocated Memory MB | Re... |
|--------------------------------|------|-------------------------|------------------|---------|----------------------|--------------------------------|--------------------------------|--------------------------------|----------|-------------|--------------------|----------------------|---------------------|-------|
| application_1790006778350_0006 | root | MedTrack Analytics - P5 | MAPREDUCE        | default | 0                    | Mon Sep 21 23:10:40 +0550 2026 | Mon Sep 21 23:10:40 +0550 2026 | Mon Sep 21 23:10:53 +0550 2026 | FINISHED | SUCCEEDED   | N/A                | N/A                  | N/A                 | N/A   |
| application_1790006778350_0005 | root | MedTrack Analytics - P4 | MAPREDUCE        | default | 0                    | Mon Sep 21 23:07:45 +0550 2026 | Mon Sep 21 23:07:45 +0550 2026 | Mon Sep 21 23:07:59 +0550 2026 | FINISHED | SUCCEEDED   | N/A                | N/A                  | N/A                 | N/A   |
| application_1790006778350_0004 | root | MedTrack Analytics - P1 | MAPREDUCE        | default | 0                    | Mon Sep 21 23:03:26 +0550 2026 | Mon Sep 21 23:03:27 +0550 2026 | Mon Sep 21 23:03:40 +0550 2026 | FINISHED | SUCCEEDED   | N/A                | N/A                  | N/A                 | N/A   |
| application_1790006778350_0003 | root | MedTrack Analytics - P2 | MAPREDUCE        | default | 0                    | Mon Sep 21 22:56:23 +0550 2026 | Mon Sep 21 22:56:24 +0550 2026 | Mon Sep 21 22:56:39 +0550 2026 | FINISHED | SUCCEEDED   | N/A                | N/A                  | N/A                 | N/A   |
| application_1790006778350_0002 | root | MedTrack Analytics - P3 | MAPREDUCE        | default | 0                    | Mon Sep 21 22:19:01 +0550 2026 | Mon Sep 21 22:19:03 +0550 2026 | Mon Sep 21 22:19:14 +0550 2026 | FINISHED | SUCCEEDED   | N/A                | N/A                  | N/A                 | N/A   |